

2(a)	any two from: (each haem contains) an, iron ion / ferrous ion / Fe^{2+}; A iron / iron atom (each haem) binds to / AW, an oxygen molecule / two oxygen atoms ; A forms a bond <i>only if</i>, iron ion / ferrous ion / Fe^{2+}, stated	2
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Question	Answer	Marks
2(b)	any three from: less carbon dioxide diffuses into red blood cells ; less carbon dioxide reacts with water ; less carbonic acid is formed ; less, dissociation of carbonic acid / formation of hydrogen ions and hydrogen carbonate ions ; less haemoglobinic acid is formed ; A description e.g. H^+ binds to haemoglobin less oxygen (released from haem) diffuses out of red blood cells (to respiring tissues) ; ref. to carbaminohaemoglobin ; AVP ; e.g. ref. to change in shape of haemoglobin when H^+ binds	3
2(c)(i)	idea that blood vessel X carries (oxygenated blood) at high pressure (to body tissues) ; A ref. to systemic circulation any two from: difference high proportion of elastic fibres (and low proportion of smooth muscle) ; explanation (elastic fibres allow X to) expand / stretch, qualified ; e.g. to accommodate / AW, blood from ventricular, systole / contraction with, surges / pulses, of blood leaving heart to prevent, bursting / rupture recoil of elastic fibres, detail ; e.g. to allow surges / pulses, to continue idea of allowing continued flow of blood to tissues maintains, onward pressure / high pressure of blood if no marks gained in this section allow one mark for: stretch and recoil with one detail or ref. to two correct details but stretch and recoil not stated	3

Question	Answer	Marks
2(c)(ii)	<i>allow, impulses / wave of excitation, for electrical impulses R nerve impulses once and allow ecf must attempt AVN <u>and</u> Purkyne tissue to gain max</i> <i>any four from:</i> <i>both involved in, co-ordination / control, of, systole / contraction, of, heart / heart muscle / cardiac muscle / ventricles ;</i> <i>AVN</i> <i>only pathway for impulses to travel from atria to ventricles ; AW</i> <i>A impulses must pass through AVN for ventricular, contraction / systole</i> <i>allow, a delay / 0.1 s, for impulse to pass from atria to ventricles ;</i> <i>so, ventricles contract after atria contract / ventricular systole occurs after atrial systole ;</i> <i>allows atria to (fully) empty before, ventricles (begin to) contract / ventricular systole</i> <i>or</i> <i>allows blood to fill ventricles before ventricular, contraction / systole ;</i> <i>Purkyne tissue</i> <i>carries / transmits / AW, impulse(s), (down septum), to base of ventricles / through walls of the ventricles ;</i> <i>for ventricular systole / AW ;</i> <i>ventricular muscles contract / AW, together / at same time / simultaneously / from the base upwards / from apex of heart ;</i>	4